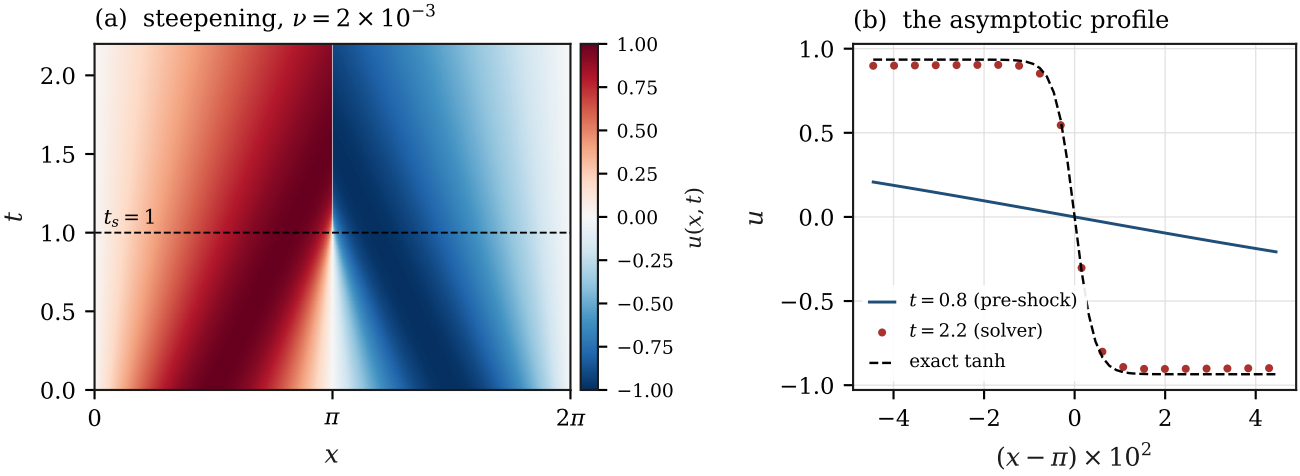
(a) steepening, $\nu = 2 \times 10^{-3}$
$t_s = 1$
t
x
$u(x,t)$
0
π
2π
0.0
0.5
1.0
1.5
2.0
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
(b) the asymptotic profile
u
$(x-\pi) \times 10^2$
$t = 0.8$ (pre-shock)
$t = 2.2$ (solver)
exact tanh
1.0
0.5
0.0
−0.5
−1.0
−4
−2
0
2
4